

Atmospheric absorption and emission

Part 2

Topics Covered

- Atmospheric energy states (cont.)
 - Vibrational energy
 - electronic energy
 - synthesis of energy states
- Broadening mechanisms w/ emphasis on pressure broadening

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Absorption and emission by gases (cont. from prev. lecture)

- Recall that this is a very high-level discussion of absorption and emission by atmospheric gases

→ Students interested in a deeper dive should look into

12.315 / 12.815

- Recall from prev. lecture:

Total energy U of isolated molecule consists of 3 energy states

$$U = U_e + U_v + U_r$$

$U_e \equiv$ electronic energy $\rightarrow$ electron energy levels in individual atoms

$U_v \equiv$ vibrational energy $\rightarrow$ displacement between 2 or more nuclei

$U_r \equiv$ rotational energy $\rightarrow$ rotation of molecule

* Key point: energy states are quantized

- Rotational energy covered last lecture:

$$U_r = \frac{\omega \hbar \sqrt{l(l+1)}}{2}$$

$\hbar \equiv$ (reduced) Planck const

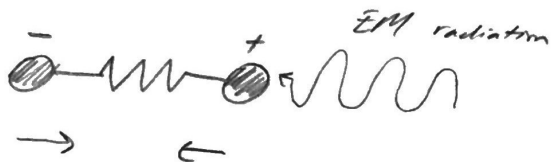
$l \equiv$ nonnegative integer "rotational quantum number"

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Vibrational energy UV

- Distance between atomic nuclei is influenced by absorption and emission of EM radiation

- Basic process



positive side of dipole is pushed by EM radiation and negative side is pulled, causing nuclei to move closer to one another



nuclei are both positively charged $\Rightarrow$ repel

$\Rightarrow$ EM radiation pushes nuclei together, like charges repel $\Rightarrow$ vibration

- Vibration frequencies are quantized $\rightarrow$ "vibration levels"

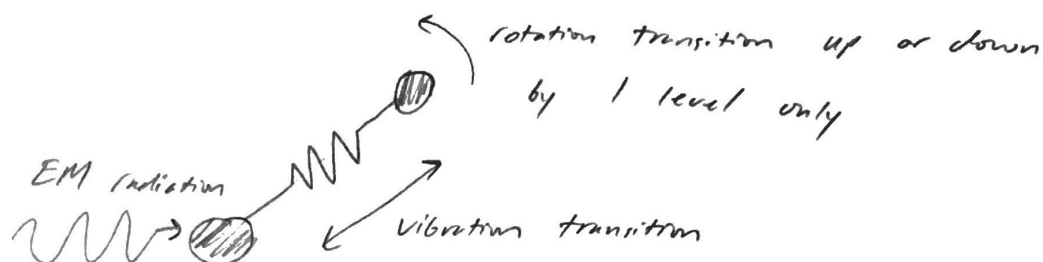
- When molecule absorbs EM radiation w/ frequency associated w/ higher vibration level, "vibration transition" occurs

- Vibration transitions involve higher energy levels than rotation transitions $\Rightarrow$ vibrational transitions are always accompanied by rotation transitions

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- Recall from last lecture that rotation transitions involve jumps of one and only one level (up or down)



- Since each vibration level can be associated w/several rotation (sub-) levels, there are many possible vibration transitions within a population
- All vibration transitions require slightly different energy levels, which results in a series of closely spaced absorption lines that converge at short wavelength (i.e. higher freq) $\Rightarrow$ absorption bands in the intermediate infrared spectrum
- Details are well beyond scope of the course, and we will simply state that the energy levels for vibrations is

$$U_V = h\omega_K \left(\nu_K + \frac{1}{2} \right) \quad \text{where } \nu_K \equiv (\text{integer}) \text{ vibrational quantum number for } K \text{ normal mode}$$

(normal mode: components of system moving sinusoidally w/same freq and relative phase)

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- Different types of molecules have different normal modes, sometimes called fundamentals

- To illustrate, consider composition of Earth's atmosphere

$N_2 \sim 78.1\%$

$O_2 \sim 20.9\%$

$Ar \sim 0.9\%$

$CO_2 \sim 0.04\%$

other (H_2 , Ne , He , CH_4 , ...)

Vibrational normal modes

	ν_1 symmetric	ν_2 bending	ν_3 antisymmetric
diatomic N_2, O_2, CO		N/A	N/A
linear triatomic CO_2, N_2O		 same energy, different quantum numbers	
bent (isosceles) triatomic			

(see slides for movie of $\rightarrow$ Ar H_2O)

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• Electronic energy U_e

- Represents electron energy levels within molecules
- Moving from one energy level to another requires absorbing or emitting a specific frequency of EM radiation

$$\Delta E = h\nu \quad (\text{Bohr's formula})$$

- Bohr postulated the quantum theory based on the observed hydrogen spectrum and by building on Planck's oscillator concept

↳ Bohr postulated that the angular momentum L of an electron could only take discrete values

$$L = n\hbar \quad n \equiv \text{principal quantum number}$$

↳ From this selection rule, Bohr showed

$$U_e \propto \frac{1}{n^2} \quad n = 1, 2, 3, \dots$$

where the constant of proportionality is a universal constant

$$U_e = -\frac{m_e e^4}{8\epsilon_0 h^2 n^2} = -\frac{R_H h c}{n^2}$$

$m_e \equiv \text{mass of electron}$

$e \equiv \text{charge of electron}$

$h = 2\pi\hbar$

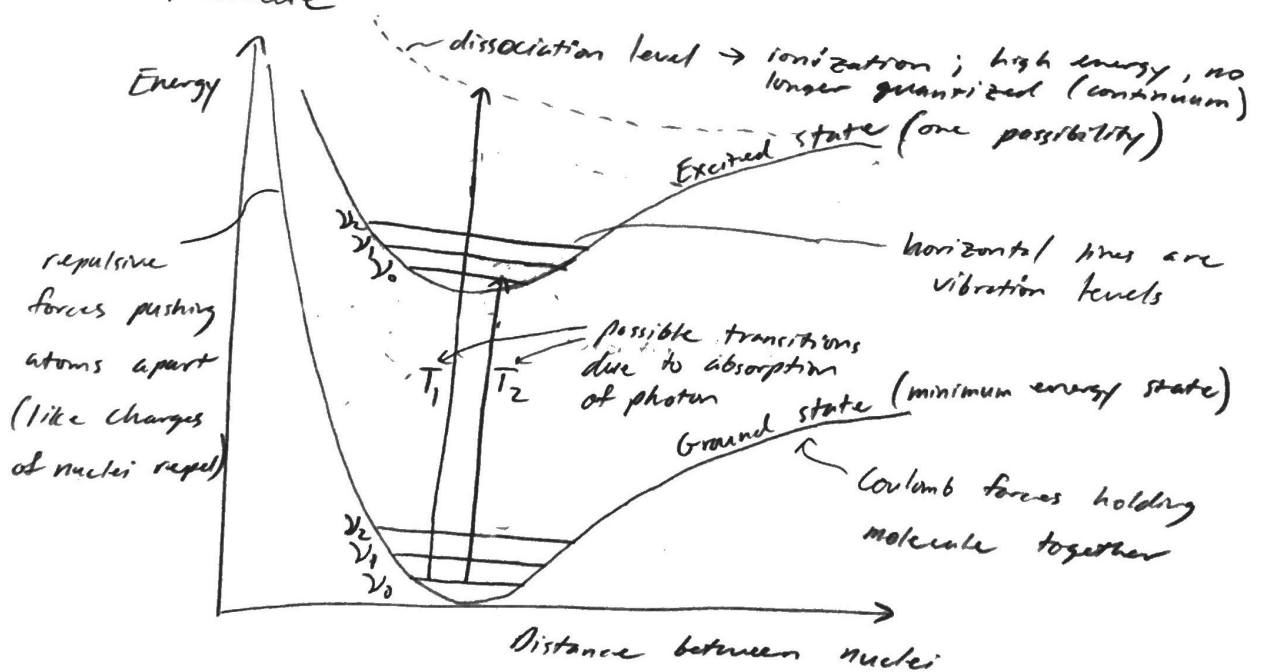
$\epsilon_0 \equiv \text{permittivity of free space}$

$R_H \equiv \text{Rydberg constant for hydrogen}$

- Putting it all together

(See Lion "An Introduction to Atmospheric Radiation" Ch 1-3 for more details)

- Electronic and vibrational energy are closely related because both come from valence bonds that bind atoms into molecules
- Sign and magnitude of force between two atoms in a molecule depends on 2 factors:
 - distance between nuclei
 - electron energy level
- To illustrate, let us consider a simple potential energy curve for two electronic states and multiple possible vibration levels of a diatomic molecule



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- Key points from potential curve

1. Energy gap from ground state to excited state (due to electronic transition) is simply the difference in minimum potential energies
2. Vibrations are represented as horizontal lines showing changes in distances between nuclei
 ↳ vibration transitions (always accompanied by rotation transitions) entail change in energy level $\Rightarrow$ range of oscillation
3. Illustrated transitions T_1 and T_2
 - T_1 from ground state to dissociation, which is not quantized $\Rightarrow$ a continuum of absorption wavelengths near energy associated w/ T_1
 - T_2 remains quantized $\Rightarrow$ transition to specific vibration level $\Rightarrow$ absorption spectrum will have discrete wavelengths in absence of disturbances

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- Key points (cont.)

4. At atmospheric temperatures, most molecules are in ground vibrational states
 $\Rightarrow$ transitions T_1 and T_2 are representative of transitions in atmosphere

5. Most absorption processes in atmosphere are a combination of electronic and vibrational transitions as depicted

$\hookrightarrow$ These generally occur in UV, visible, and near IR frequencies

• Effects of disturbances on absorption (and emission) spectrum

- Previous discussion applicable to isolated, undisturbed, stationary molecular system

- But molecules are in constant motion, interacting and colliding w/ one another and w/ other matter (e.g., dust particles)

$\Rightarrow$ Energy levels during transitions change slightly due to external influences on molecules $\Rightarrow$ radiation emitted during repeated transitions is polychromatic (multiple frequencies), causing absorption spectra lines of finite width

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- Increased absorption width called "line broadening"
- Two main causes of line broadening
 1. Collision or pressure broadening, which is dominant in troposphere
 2. Doppler broadening - results from different thermal velocities of atoms and molecules (Doppler effect) $\rightarrow$ only important in upper atmosphere where pressures are very low
- Pressure broadening is, in general, the dominant process influencing absorption and emission in Earth's atmosphere
- We can understand pressure broadening by going back to our discussion of permittivity in Lecture 1 where we modeled polarization of materials as a damped harmonic oscillator. Same model applies for pressure broadening

optical properties
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- Therefore, dielectric constant is

$$\epsilon(\omega) = 1 + \frac{\omega_p^2}{(\omega_0^2 - \omega^2) + j\gamma\omega}$$

where

$$\omega_p \equiv \text{plasma frequency} = \frac{Ne^2}{\epsilon_0 m_e}$$

$e \equiv$ electron charge

$N \equiv$ number of dipoles

$m_e \equiv$ mass of electron

$\epsilon_0 \equiv$ permittivity of free space

$\omega_0 \equiv$ natural frequency

$\gamma \equiv$ damping coef.

- Defining the power absorption coef as we did for volume scattering, we have

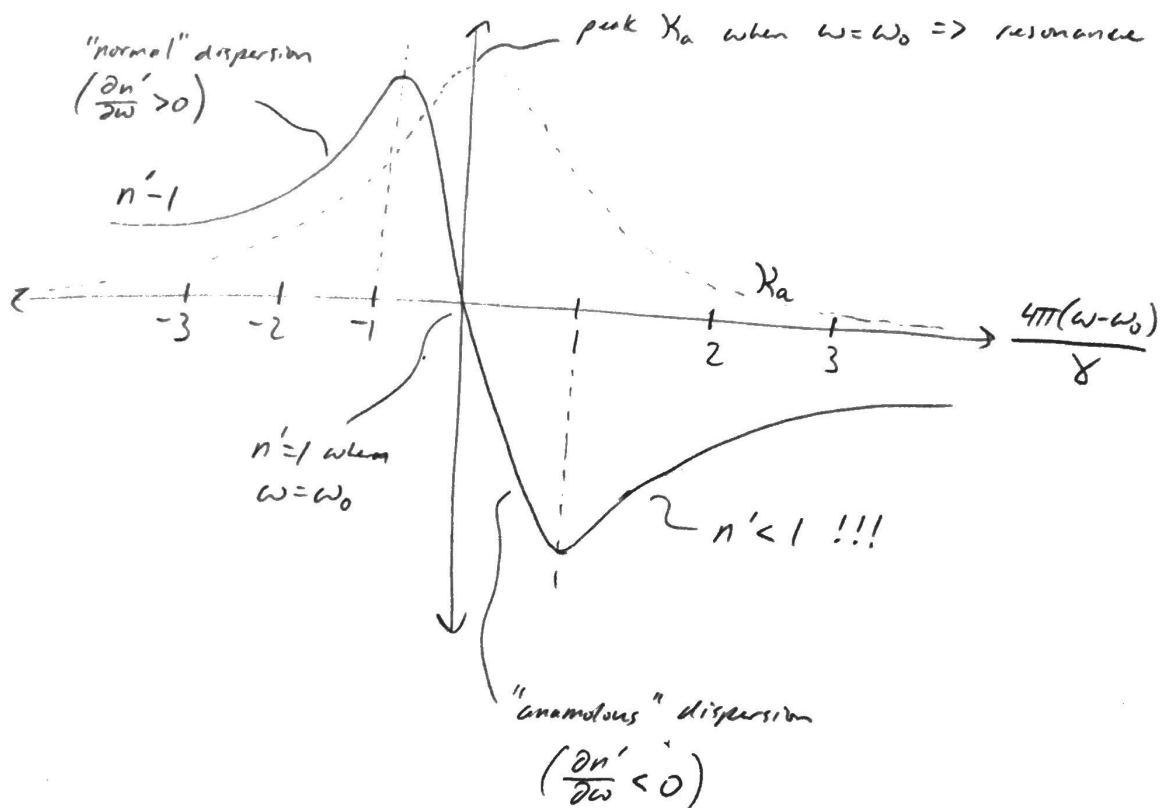
$$\begin{aligned} \chi_a &= 2K_0 n'' = 2K_0 \text{Im}\{\sqrt{\epsilon}\} \\ &= \frac{S\gamma}{(\omega - \omega_0)^2 + \gamma^2} \rightarrow \text{sometimes called the Lorentz profile} \end{aligned}$$

where $S = \frac{\omega_p^2}{c} \equiv$ spectral line strength

- Similarly, the real component of the index of refraction is

$$n' = 1 + \frac{\omega_p^2}{\omega_0} \frac{(\omega - \omega_0)}{(\omega - \omega_0)^2 + \gamma^2}$$

- Pressure broadening (cont.)

Plotting χ_a and n' - Take-aways from χ_a and n' plot:

- For χ_a : 1) Peak absorption at resonance freq ω_0
- 2) width of χ_a defined by damping coeff γ

→ what causes damping? emission
(no apparent dissipative mechanisms)

But collisions play a role → collision of molecules causes phase shifts in emitted EM wave. Collisions → random ⇒ population of molecules emit EM waves w/ random phases
⇒ reduced phase coherence ⇒ damping

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- Since molecular collision plays a key role in damping, we may suppose a relationship between damping coefficient and temperature and pressure

$$\gamma = \gamma_0 \left(\frac{P}{P_0} \right) \left(\frac{T_0}{T} \right)^{\frac{1}{2}} \quad \text{for } \frac{1}{2} \leq \gamma \leq 1$$

$\gamma_0 \equiv$ reference damping coefficient ($\gamma = \gamma_0$ for $P = P_0$ and $T = T_0$)

$P_0 \equiv$ atmospheric pressure at sea level

$$T_0 = 273 \text{ K}$$

Plugging in temperature lapse rate for troposphere and P profile (see last lecture)

$$\frac{\gamma}{\gamma_0} = e^{-z/H_p} \left(1 - \frac{a z}{T_0} \right)^{-\frac{1}{2}} \quad \leftarrow \text{note sign}$$

$$H_p \approx 8.4 \text{ km}$$

$$a \approx 6.5 \text{ K/km}$$

$\Rightarrow \gamma$ decreases by roughly $1/e$ from surface to tropopause ($\approx 12 \text{ km}$ altitude)

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• For n' :

1. For most ω $\frac{\partial n'}{\partial \omega} > 0 \rightarrow$ "normal" dispersion

(recall phase velocity $v_p = \frac{\omega}{k_0 n'} = \frac{c}{n'}$)

but near resonance freq ω_0 , $\frac{\partial n'}{\partial \omega} < 0$

$\rightarrow$ "anomalous" dispersion $\rightarrow$ only occurs where k_a is large

and

$$\frac{\partial n'}{\partial \omega} = 0 \quad \text{when} \quad \omega = \omega_0 + \frac{\delta}{4\pi}$$

2. $n' = 1$ when $\omega = \omega_n$

3. strikingly, $n' < 1$ when $\omega > \omega_0$ (in vicinity of ω_0 , not in general)

This implies $v_p > c$!!! called

"superluminal" $\rightarrow$ many of you noticed this in question 1 of pset 1

Is it physical? Yes

Explanation:

- n' varies $\omega/\omega \Rightarrow$ atmosphere is dispersive, at least in the neighbourhood of ω_0
- phase velocity only tells us how fast a wave of a given frequency travels (i.e., how fast a plane of constant phase travels)

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- In a dispersive material, waves of different frequencies travel at different speeds
 $\Rightarrow$ wave packet distorts



- Useful then to define the rate of propagation of a plane of constant amplitude
 $\rightarrow$ group velocity $V_g = \frac{\partial \omega}{\partial k}$
- Energy (information) travels at the group velocity
 $\Rightarrow$ no requirement that $V_p \leq c$ even though $V_p < c$ is by far the most common situation
 (most of you will never encounter a case where $V_p > c$ but this example is useful to highlight $V_g = \frac{\partial \omega}{\partial k}$, an extremely useful concept that some of you will encounter in practice)

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• Doppler broadening

- Pressure broadening is the dominant process in much of the atmosphere, but in the upper atmosphere (stratosphere and beyond) pressures are low $\rightarrow$ pressures decrease w/ altitude while temperature increases w/ altitude $\Rightarrow$ damping due to pressure broadening decreases rapidly w/ altitude $\Rightarrow$ in stratosphere and above, pressure broadening is negligible and doppler broadening becomes dominant
- From the kinetic theory of gases and assuming thermodynamic equilibrium, it can be shown that the absorption coefficient for doppler broadening is

$$K_d = \frac{S}{K_0 \sqrt{\pi}} \exp \left\{ - \left(\frac{K - K_0}{2\pi K_0} \right)^2 \right\}$$

where

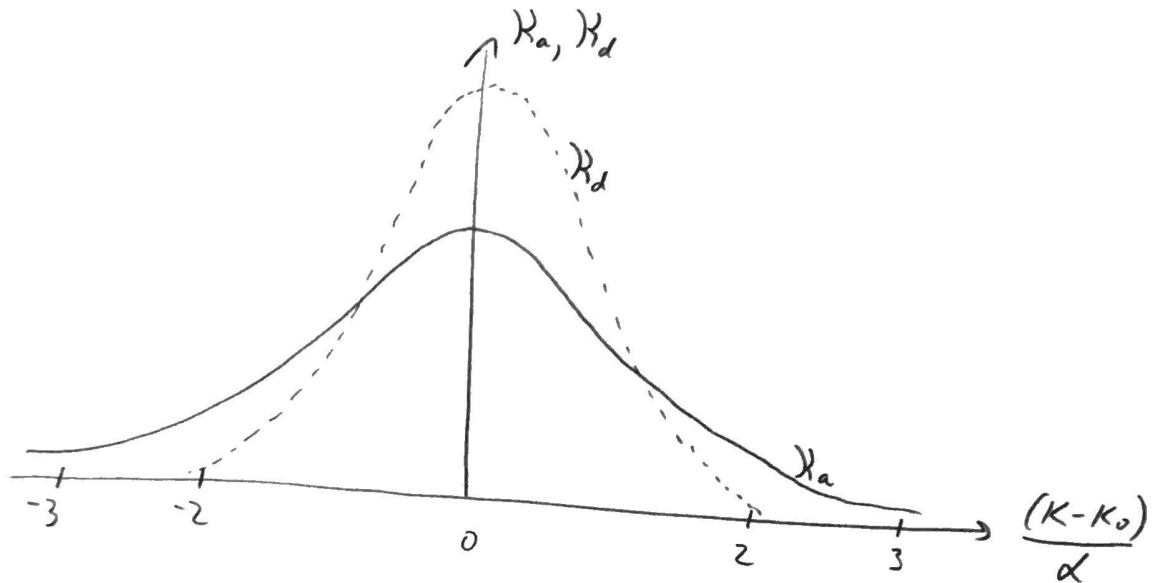
$$K_0 = \frac{K_b}{2\pi} \sqrt{\frac{2 K_B T}{m_m c^2}}$$

$K_B \equiv$ Boltzmann's constant

$m_m \equiv$ molecular mass

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Comparing pressure and doppler broadening



where $\alpha = \begin{cases} \frac{\delta}{4\pi} & \text{Pressure broadening} \\ \alpha_D & \text{Doppler broadening} \end{cases}$

Brief summary:

- Frequencies of absorption bands are determined by atmospheric composition and the mechanisms of molecular rotation, vibration, and electronic levels. These are in order of increasing energy $\Rightarrow$ vibration and electronic transitions are dominant and tend to manifest and cluster at higher frequencies (near IR + higher freq.)
- Other mechanisms, namely collisions of molecules, broaden these absorption bands $\rightarrow$
- Together gives rise to the atmospheric transparency we've seen in class